



THE UNIVERSITY OF
NOTRE DAME
A U S T R A L I A

MED 4000

WEEK 20

TUTOR GUIDE

CASE ONE

Short case number: 4_20_01

Category: Immune & Haemopoetic Systems
Infectious Diseases

Discipline: Medicine

Setting: General Practice & Hospital outpatient clinic_Urban

Topic: HIV & AIDS (long case with opportunistic infection)

CASE

Chris Booker is a 28 year old accountant who presents to you for hepatitis C testing. He was concerned when someone he knew developed Hepatitis C, and discloses that he had a brief period of experimentation with injecting drug use (IDU) 3 years ago. During that time he had shared needles on a couple of occasions. Nowadays he occasionally uses marijuana but has not injected since. Chris has no other risk factors or significant past history.

QUESTIONS

First presentation:

1. You realise that IDU is also a risk factor for HIV infection and you suggest to Chris that he should also have an HIV test. What are the other indications for HIV testing? Can you order the test without him knowing?
2. What are the risk factors known to be associated with HIV infection in Australia? What are the differences in patterns of transmission between indigenous and non-indigenous population?
3. He agrees to your recommendation and returns two weeks later for his results. The Hepatitis C test was negative but the HIV test was positive. What points will you cover in post-test counselling?
4. Chris is shocked and distressed at receiving the HIV result. You ask him to come back in a week to go through the immune function tests and discuss the treatment options open to him. He says that his girlfriend will think that he is gay. Can she ring up and speak to you about his result? What are the professional and legal considerations regarding confidentiality?
5. At the visit when he received his HIV test result, Chris had FBC and immune function tests taken, but neither Chris nor his girlfriend show up for the follow up consultation. You are unable to reach him despite numerous attempts. What are your responsibilities regarding follow up with Chris?

CASE [Con't]



b) Two years later Chris presents with recent onset of loss of visual acuity and floaters in his right eye. You are now seeing him in Ophthalmology outpatients.

On examination, he is afebrile and has no abnormal physical findings apart from abnormal findings in the right eye on indirect ophthalmoscopy consisting of granular yellowish lesions following a vascular distribution with perivascular exudates and haemorrhages.

His current blood results are:

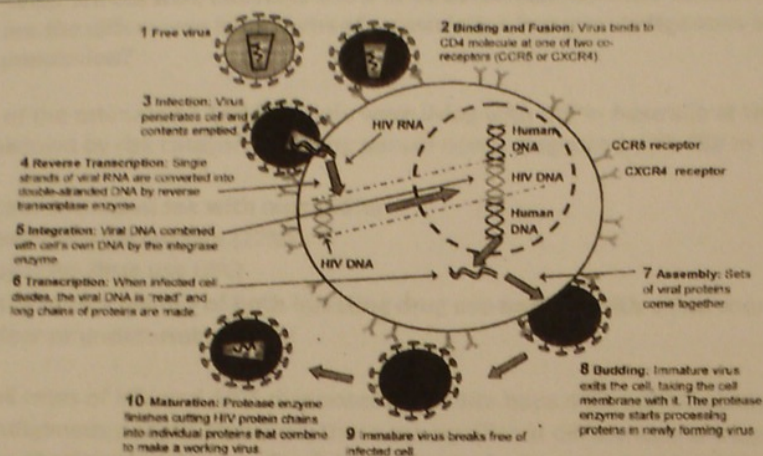
Haemoglobin of 12.6 g/dl,
white cell count $2.1 \times 10^9/l$,
platelet count of $150 \times 10^9/l$,
CD4+ cell count of 100 cells/mm³.

[FBC from 2 years ago: Haemoglobin of 14.5 g/dl, white cell count $4.3 \times 10^9/l$, platelet count of $250 \times 10^9/l$, CD4+ cell count of 350 cells/mm³.]

FURTHER QUESTIONS

The Senior Ophthalmology registrar confirms CMV retinitis.

6. In addition to full history and examination, what other tests should be carried out to assist in management of this patient?
7. List the opportunistic infections (OIs) associated with HIV infection and at what point of immunosuppression they tend to occur. For a treatment-naïve patient like Chris, who presents with an OI, what is the approach to management of the OI and underlying HIV infection?
8. Using the figure below, outline the broad categories of drugs used in the treatment of HIV in relation to their mechanisms of action. Describe the limitations to their use with compliance, adverse drug reactions, drug interactions, co-morbidities and resistance.



9. Despite his initial reluctance, you finally convince him to commit to life-long anti-retroviral treatment. Which of the indications for initiating anti-retroviral therapy does Chris satisfy?

Resources:

- Kumar P & Clark M. Clinical Medicine, 7th edition. Chapter 4 Infectious diseases, tropical medicine and sexually transmitted infection. Pp 184-201. Saunders Elsevier 2009.
- Wilkins EGL. Human immunodeficiency virus infection and the human acquired immunodeficiency syndrome. Chap 14, Davidson's Principles and Practice of Medicine 20th Edition. Churchill Livingstone Elsevier 2006. 377-402.
- Rogers G. How to treat: HIV care and prevention – part 2, Aust Doctor Feb 2006, pp 27-34.
- ASHM Guide to HIV Laws and Policies for Healthcare Professionals
<http://www.ashm.org.au/HIVLegal/Default.asp?PublicationID=2>
- Guidelines for the use of Antiretroviral Agents in HIV-1-Infected Adults and Adolescents April 2010. Developed by the DHHS Panel on Antiretroviral Guidelines for Adults and Adolescents (USA) with the Australian Antiretroviral Guidelines Panel providing commentary to adapt the guidelines to the Australian setting.
- The Australasian Contact Tracing Manual (Edition 4, Oct 2010)
<http://www.ashm.org.au/images/publications/aust-contact-tracing.pdf>

ANSWERS

1. You realise that IDU is a risk factor for HIV infection and you suggest to Chris that he should also have an HIV test. What are the other indications for HIV testing? Can you order the test without him knowing?

Indications for HIV testing should be assessed on the basis of the following risk factors:

- unprotected male-to-male intercourse
- sharing of injecting equipment
- being the sexual partner of an HIV positive person
- being from a country with a high HIV prevalence
- having recently travelled overseas
- presenting for post-exposure prophylaxis (PEP) after occupational/non-occ exposure
- pregnancy
- requesting an HIV test in the absence of clear risk factors
- diagnosis of a sexually transmissible infection

Can you order the test without him knowing? NO. The Medicare Benefits Schedule (MBS) stipulates that a practitioner requesting an HIV test must ensure that the patient has given informed consent, received adequate pre-test discussion, and understands that further discussion may be necessary after the test results are received. Some States and Territories have specific legal regulations relating to pre-test and post-test discussions for HIV testing.

When a person requests or is offered an HIV test, the practitioner should assess a person's preparedness to be tested and give appropriate information about risk, points of referral if necessary, assurances about confidentiality and privacy. The information should be given in a way that is appropriate to the person's gender, culture, behaviour and language. Under the National HIV Testing Policy, the pre-test discussion should include:

- Information on how HIV is transmitted (where appropriate)
- Risk assessment and discussion of the reason for testing
- Timing of the risk event and options for PEP
- Possible desirability of other sexually transmissible infections (STIs) testing
- Information about confidentiality and privacy
- Information about the testing process including how results are to be provided, the window period, and the difference between HIV and AIDS
- Information about what happens to test results
- Seeking informed consent for the test to be conducted
- Assessment of the person's preparedness to be tested and assurance that the person wishes to proceed with the test
- Information about what a negative or positive result means
- Assessment of support mechanisms while waiting for the test result and if the result is positive.

2. What are the RISK FACTORS known to be associated with HIV infection in Australia? What are the differences in patterns of transmission between indigenous and non-indigenous population?

Out of the estimated 17,444¹ people were living with HIV in Australia at the end of 2008. The breakdown by risk categories among people newly diagnosed with HIV in 2008 is as follows:

- men who have sex with men (66%)
- heterosexual contact (27%)
- injecting drug use (3%)
- men with a history of both injecting drug use and sex with other men (3%)
- other or undetermined (5%)

Overall rates of HIV and AIDS diagnoses per capita have differed little between indigenous and non-indigenous people. However, there are significant differences in transmission routes. Among Aboriginal and Torres Strait Islander people diagnosed with HIV in 2004-2008, sex between men was the reported source of exposure to HIV in 54% of the population (compared to 66% in the non-Indigenous population). Heterosexual contact was the reported transmission route for 23% of cases in both the Aboriginal and Torres Strait Islander population and the non-Indigenous population. The number of cases attributable to injecting drug use was higher among the Indigenous population (22%) than the non-Indigenous (3%). Women accounted for a higher proportion of total HIV infections among the Aboriginal and Torres Strait Islander population (26.9%) than among the non-Indigenous population (11.6%).

3. He agrees to your recommendation and returns two weeks later for his results. The Hepatitis C test was negative but the HIV test was positive. What points will you cover in post-test counselling?

The National HIV Testing Policy states that **post-test discussion following an HIV-positive test result (which must be given in person)** should include issues such as:

- immediate needs and support
- safe behaviours: education, information and support
- whom the person should tell and how, including information around the person's rights regarding disclosure
- managing or understanding strong emotions, feelings, reactions and changes, including ways to deal with loss and grief, depression, anger and anxiety
- options in drug treatments and medical management
- ongoing counselling or therapy if required
- complementary and alternative management options
- strategies for managing HIV that are flexible and appropriate to the person's needs
- legislative requirements (notification, contact tracing, storage and coding).

¹ From the start of the epidemic until the end of 2008, there have been 28,330 diagnoses of HIV and 10,348 diagnoses of AIDS. Australia has recorded 6,765 AIDS deaths.

It is important to state the result of the HIV test clearly as some people confuse a 'positive result' with a 'good' result. **Shock or fear associated with a positive HIV result will impact most people's capacity to absorb information immediately following diagnosis.** Patients usually need time to process and react to their test result and any other information they are provided during that appointment. Prompt follow-up appointments, written information and/or referral to support agencies is often necessary, particularly an after-hours support contact number. It may be useful to help the patient plan the next 24-48 hours including a follow-up appointment within that period. Some patients feel an urgent need to tell people what has happened, so it is useful to discuss the implications of disclosure and privacy to assist the patient's consideration of the implications of those actions. Of course it is necessary to explain and reinforce the patient's responsibilities to avoid transmitting HIV to others, which may lead to the contact tracing process.

Communicating bad news: Before communicating bad news, the environment and length of consultation require additional consideration, as does the patient's preference for having a person of his or her choice present. In some circumstances, it may be necessary to suggest immediate or early access to additional counselling and support services. A prompt follow-up consultation may be helpful when a condition is diagnosed that is likely to involve hospitalisation, sustained treatment and lifestyle change, or risk of permanent impairment or death. This will give the patient an opportunity to absorb the information, and to think of questions he or she may wish to ask.

From: *Communicating with Patients: Advice for Medical Practitioners* (Australian Government: National Health and Medical Research Council 2004)

4. **Chris is shocked and distressed at receiving the HIV result. You ask him to come back in a week to go through the immune function tests and discuss the treatment options open to him. He says that his girlfriend will think that he is gay. Can she ring up and speak to you about his result? What are the professional and legal considerations regarding confidentiality?**

Even though Chris has given permission, it is not advisable to provide information about his HIV status over the telephone because it restricts your ability to verify the identity of the person to whom you are giving this information. So you should recommend that his girlfriend to attend with him at the next consultation so that her queries and concerns can be addressed in his presence.

It is important to maintain privacy and confidentiality because:

- patients are concerned about the stigma and discrimination associated with their HIV and related conditions
- patients want to know that they can choose who has access to information about them
- patients are far more likely to seek medical care and give full and honest accounts of their symptoms if they feel comfortable, respected and secure
- a health system with strong privacy mechanisms will promote public confidence and trust in health care services generally.

Health care workers must not breach a person's privacy or confidentiality except in a very limited number of circumstances and practitioners are advised to seek further clarification of the situation in their jurisdiction as definition of 'professional' in cases of needlestick injury is not uniform Australia-wide. Exemptions are broadly described as:

- communicating information to others directly involved in the treatment of a patient during a particular episode of care
- cases of needle-stick injury where a professional is aware of a patient's positive status and a health care worker has been exposed in circumstances where there is a real risk of transmission and it is not possible to conceal the identity of the source patient who has refused to consent to disclosure
- provision of medical services in a particular instance of care where there is a need to know the infection status for treatment purposes of benefit to the patient (e.g. in an emergency or if the patient is unconscious). This should not, however, detract from the observance of standard infection-control precautions.

The exceptions to confidentiality of HIV result in a **LEGAL** context are: mandatory notification and when a patient may put others at risk.

HIV is a notifiable disease in all Australian states and territories. Legal obligations around notification are mandated by state laws, which define a doctor's duty to notify the respective health department of a notifiable disease. Specific state laws also allow a health care practitioner to notify their health department of a patient's name and other confidential information **where they fear that patient may be putting others at risk**. Where the law requires disclosure of confidential information, there can be no action for breach of confidence.

The National Guidelines provide a public health framework based upon 5 levels of intervention. The five recommended levels of intervention are:

Level 1: an identified individual should be provided with comprehensive counselling, education and support.

Level 2: if it is clear that the initial intervention will not lead to the necessary change in behaviour, then the National Guidelines provide for the introduction of an HIV Advisory Panel to provide formal support to the individual's primary health care provider.

Level 3: the Chief Health Officer, on advice from the Advisory Panel, may make a behavioural order which requires the individual to undergo counselling, treatment, refrain from specific behaviour or be subject to supervision.

Level 4: the Chief Health Officer, again on advice from the Advisory Panel, may make an order for the person to be subject to isolation or detention.

Level 5: referral of the individual to police.

5. Neither Chris nor his girlfriend attend for the follow up consultation. You are unable to reach him despite numerous attempts. What are your responsibilities regarding follow up with Chris?

Given the case precedents cited below, it is essential for you as the practitioner to document fully the discussions with Chris and the advice given to him regarding the diagnosis, treatment options, counselling and other information as well as his obligations to sexual partner(s). Any attempts made to contact him for follow-up and mode of contact must be also documented fully and dated.

There has been very little case law relating to legal obligations around contact tracing. A summary of key Australian case law is outlined:

- **Duty of care to a sexual partner** (NSW case, *BT v Oei* [1999] NSWSC 082) requires a doctor who believes a patient is HIV-infected to inform that patient of the danger he/she poses to others and to advise the measures he/she should take to protect others from cross-infection
- **Failure to follow-up positive test results** (*Kite v Malycha* (1998) 71 SASR 321). Because testing for HIV antibody status is largely anonymous (that is, the blood sample is often sent without patient identification but with a code assigned by the referring doctor), this judgement makes it probable that a doctor who does not have a system to check that results are received for all requested/referred tests and that those results are communicated to the relevant patient, will be held to be negligent.
- **Counselling obligations and duty of care to a sexual partner** (*PD v Dr Y & Dr X* [2003] NSWSC 487). It would not have been necessary for the doctors to breach their common law duty of confidentiality to the man to prevent foreseeable harm to the woman, because they had a statutory duty to report the man to the health department, officers of which would have performed the necessary contact tracing and notification. **Many of the judge's findings were based on the inadequacy or absence of Dr X's clinical records.** Dr X was found to be vicariously liable for the omissions of Dr Y. Had the notes been cross-referenced, the doctors may have realised the full extent of the situation and contacted the patient before she contracted HIV.

6. The Senior Ophthalmology registrar confirms CMV retinitis. You are able to access his old notes and notice that he had a FBC taken 2 years ago but no other tests. In addition to full history and examination, what other tests should be carried out to assist in management of this patient?

Baseline investigations in a newly diagnosed HIV infection

(Box 4.18 Kumar & Clark and Guidelines for the use of Antiretroviral Agents April 2010)

- Haematology: Full blood count (FBC), differential count and film
- Biochemistry including liver and renal function, Fasting blood glucose and serum lipids (if the patient is considered at risk of cardiovascular disease and for baseline evaluation prior to initiation of combination antiretroviral therapy)
- Immunology: CD4 T-cell count (more notes below)
- Virology: HIV antibody, HIV viral load (more notes below on plasma HIV RNA testing), HIV genotype, Hepatitis A, B, C serology, CMV antibody (particularly in this patient)
- Microbiology: Toxoplasma serology, other STD tests incl Syphilis (RPR/VDRL and TPHA)
- Other: Cervical cytology, Chest X-ray if indicated

According to the Baseline Evaluation, the following laboratory tests should be performed for a new patient during initial patient visits:

- **Plasma HIV RNA (viral load)**
- Resistance testing² should be used to guide selection of an antiretroviral regimen in both treatment-naïve and treatment-experienced patients.

Note: CD4 T-cell count and plasma HIV RNA are 2 surrogate markers are used routinely to assess the immune function and level of HIV viremia.

CD4+ T-CELL COUNT

The CD4+ T-cell count (or CD4 count) serves as the major clinical indicator of immune function in patients who have HIV infection. It is one of the key factors in deciding whether to initiate antiretroviral therapy and chemoprophylaxis for opportunistic infections, and is the strongest predictor of subsequent disease progression and survival. A significant change (2 SD) between 2 tests is approximately a 30% change in the absolute count or an increase or decrease in CD4 percentage by 3 % points.

- **Use of CD4 Count for Initial Assessment:** The CD4 count is one of the most important factors in the decision to initiate antiretroviral therapy and/or prophylaxis for opportunistic infections.
- **Use of CD4 Count for Monitoring Therapeutic Response:** An adequate CD4 response for most patients on therapy is defined as an increase in CD4 count in the range of 50–150 cells/mm³ per year, generally with an accelerated response in the first 3 months and an average increase of approximately 50–100 cells/mm³ per year for the subsequent years.

PLASMA HIV RNA TESTING

Viral load testing serves as a surrogate marker for treatment response and can be useful in predicting clinical progression. The minimal change in viral load considered to be statistically significant (2 SDs) is a threefold, or a 0.5 log₁₀ copies/mL change. One key goal of therapy is suppression of viral load to below the limits of detection (< 40 copies/mL by most commercially available assays).

For most individuals who are adherent to their antiretroviral regimens and who do not harbor resistance mutations to the prescribed drugs, viral suppression is generally achieved in 12–24 weeks, even though it may take a longer time in some patients.

7. List the opportunistic infections (OIs) associated with HIV infection and at what point of immunosuppression they tend to occur. For a treatment-naïve patient like Chris, who presents with an OI, what is the approach to management of the OI and underlying HIV infection?

Correlations between CD4 count and Opportunistic Infections (Table 14.7, Ob 384 Davidson's)

CD4 count	Opportunistic Infection (OI)
>500 (cells/mm ³)	- Recurrent vaginal candidiasis

² Resistance testing is recommended for:

- Treatment-naïve patients at initiation of Rx
- Virologic failure (HIV RNA >1000 copies/mL) when changing therapy
- Suboptimal viral load reduction.

<500 (cells/mm ³)	- Kaposi's sarcoma - Pulmonary tuberculosis - Oropharyngeal candidiasis - Herpes zoster - Pneumococcal pneumonia
CD4 count	Opportunistic Infection (OI)
<200 (cells/mm ³)	- Pneumocystis jirovecii (previously known as P. Carinii) - Mucocutaneous Herpes simplex - Cryptosporidium - Microsporidium - Oesophageal candidiasis - Extrapulmonary TB
<100 (cells/mm ³)	- Toxoplasmosis - Cryptococcal meningitis
<50 (cells/mm ³)	- CMV retinitis/GIT disease - Mycobacterium avium intracellulare

- When a patient presents with an acute OI, treatment of the OI must take precedence over management of the HIV coinfection, since there is an acute process that carries a definite early mortality if it is not managed properly.
- It is also important to consider the timing of antiretroviral therapy. Immediate antiretroviral therapy (involving concurrent treatment of the OI and HIV) raises concerns regarding additive drug toxicities, drug-drug interactions between OI therapy and ARV, adherence issues, and the potential for immune reconstitution inflammatory syndrome (IRIS)³. On the other hand, early initiation of antiretroviral therapy has the potential to both preserve and rapidly restore pathogen-specific cellular and humoral responses. It is quite possible that outcomes for an OI may be better with a more-recovered immune system.

So in this case, initiating steroid and valganciclovir treatment and continue antiretroviral therapy is the best choice because steroid treatment is necessary to reduce the patient's ocular inflammation, which may be a result of immune reconstitution inflammatory syndrome. He also needs to continue to receive antiretroviral therapy to suppress viral replication and to support immune reconstitution as much as possible.

8. Using the figure below, outline the broad categories of drugs used in the treatment of HIV in relation to their mechanisms of action. Describe the limitations to their use with compliance, adverse drug reactions, drug interactions, co-morbidities and resistance.

³ IRIS: For many Opportunistic infections, an inflammatory reaction may occur when treatment for those OIs is started concurrently with antiretroviral therapy, often contributing to the morbidity and mortality from these conditions. This paradoxical deterioration in clinical status after initiating antiretroviral therapy is attributable to the recovering immune system's response to latent/subclinical infectious or non-infectious processes and has been referred to as IRIS. It can be difficult to distinguish this presentation from progression of the original OI, a new OI, or adverse effects of therapy

Table 1. Antiretroviral drugs available in Australia

Drug	Brand names	Abbreviations
Non-nucleoside analogue reverse transcriptase inhibitors		
nevirapine	Viramune	NVP
efavirenz	Stocrin	EFV
delavirdine	Rescriptor	DLV
Nucleoside analogue reverse transcriptase inhibitors		
zidovudine	Retrovir	AZT, ZDV
didanosine	Videx	ddl
lamivudine	3TC	3TC
stavudine	Zerit	d4T
abacavir	Ziagen	ABV, 1592
emtricitabine	Emtriva	FTC
Nucleotide analogue reverse transcriptase inhibitor		
tenofovir	Viread	TFV, TDF
Protease Inhibitors		
saquinavir	Invirase	SQV
indinavir	Crixivan	IDV
ritonavir	Norvir	RTV
nelfinavir	Viracept	NFV
amprenavir	Agenerase	APV
lopinavir/ritonavir	Kaletra	KLT, LPV/r
atazanavir	Reyataz	ATZ
fosamprenavir (pro-drug of amprenavir)	Telzir	fAPV
Entry inhibitor		
enfuvirtide	Fuzeon	T-20

NB: zalcitabine/ddC (Hivid) and the Fortovase preparation of SQV are about to be withdrawn

Update on drugs listed include: New NNRTI - Etravirine (Intellece) and PIs - Darunavir (Prezista) ,
 Tipranavir (Aptivus), and 2 new classes of drugs: a) Integrase inhibitor - Raltegravir (Isentress)
 b) Fusion/Entry inhibitor - Maraviroc (Celsentri).

Table 2: Common short-term side effects of antiviral therapy

Side effect	Drugs most often associated
Nausea	zidovudine, nelfinavir, lopinavir/ritonavir
Headache	zidovudine
Increased stool frequency or frank diarrhoea	nelfinavir, saquinavir, ritonavir, amprenavir, lopinavir/ritonavir
Neurocognitive symptoms such as vivid dreams and subjective impairment	efavirenz
Peeling lips and dystrophic nails	indinavir
Ureteric stones (crystallised drug)	Indinavir
Anaemia	zidovudine
Macrocytosis	zidovudine, stavudine
Harmless hyperbilirubinaemia	indinavir, atazanavir
Raised liver enzyme levels	Nucleoside/nucleotide reverse transcriptase inhibitors, nevirapine
Hypersensitivity syndromes	sabacavir, nevirapine

Broad considerations when choosing anti-retroviral therapy (ARV):

- **Compliance:** Ease of compliance and fit of drug regimen around a patient's lifestyle
 - **Adverse drug reactions:** Can be confused with other pathology. Potential for interactions with non-HIV drugs incl herbal remedies
 - **Drug interactions:** Drug levels can be compromised by introduction of other medications
 - **Co-morbidities:** ARVs may exacerbate coexisting infections/conditions
 - **Resistance**
- Antiretroviral drugs must be taken EXACTLY as prescribed so the patient must be informed of the reasons for strict adherence. **Fixed drug combinations**⁴ have helped improve compliance. (Refer to Table 4.53, p192-3, in Kumar & Clark for list of antiretroviral drugs and adverse drug reactions.) Certain herbal remedies e.g. St John's wort induce CYP450
- Concern about possible drug interactions may favor the use of NNRTI therapy.**
- Particular care should be taken with ENZYME INDUCERS (rifampicin, rifabutin, warfarin, nevirapine) which will reduce effective levels of the ARV. Check potential interactions and consider therapeutic drug monitoring.
- Immune Reconstitution Inflammatory Syndrome (IRIS)⁵ e.g. hepatic dysfunction in co-infection with HepB/HepC, CNS symptoms with cryptococcosis.
- Resistance considerations, may favor the use of boosted Protease inhibitors**⁶.

9. Despite his initial reluctance, you finally convince him to commit to life-long anti-retroviral treatment. Which of the indications for initiating anti-retroviral therapy does Chris satisfy?

Chris should be treated with ARV therapy immediately with the one pill a day triple combination at the same time as the treatment of CMV for the reasons covered in Q. 6. Regarding initiation of ARV therapy in general, in the updated version of the Guidelines for the use of Antiretroviral Agents in HIV-1-Infected Adults and Adolescents April 2010, the Panel recommends earlier initiation of antiretroviral therapy with the following specific recommendations:

⁴ **Fixed dose combinations:**

- Atripla: Tenofovir + FTC + Efavirenz, the most commonly used combination
- Truvada: Tenofovir + FTC, the next most commonly used
- Kivexa: Abacavir + 3TC, the next most commonly used
- Combivir: AZT + 3TC, less commonly used except possibly in pregnant women.

⁵ **IRIS:** For many Opportunistic infections, an inflammatory reaction may occur when treatment for those OIs is started concurrently with antiretroviral therapy, often contributing to the morbidity and mortality from these conditions. This paradoxical deterioration in clinical status after initiating antiretroviral therapy is attributable to the recovering immune system's response to latent/subclinical infectious or non-infectious processes and has been referred to as IRIS. It can be difficult to distinguish this presentation from progression of the original OI, a new OI, or adverse effects of therapy.

⁶ **Resistance testing** (genotyping only) data from VIC & NSW in 2008 confirmed low rate for protease inhibitors (PIs): 3.3% rate to NRTI, 5.6% rate to NNRTI and consistently low for PIs.

- Antiretroviral therapy should be initiated in all patients with a history of an AIDS-defining illness or with CD4 count < 350 cells/mm³.
- Antiretroviral therapy should also be initiated, regardless of CD4 count, in patients with the following conditions: pregnancy (AI), HIV-associated nephropathy (AII), and hepatitis B virus (HBV) coinfection when treatment of HBV is indicated.

According to the Guidelines mentioned, the 4 regimens are now listed as Preferred regimens for treatment-naïve patients are:

- **efavirenz/tenofovir/emtricitabine[#]**
- ritonavir-boosted atazanavir + tenofovir/emtricitabine
- ritonavir-boosted darunavir + tenofovir/emtricitabine
- raltegravir + tenofovir/emtricitabine

[#] Note that Efavirenz plus 2 NRTIs is a possible choice that is highly effective in most patients and potentially offers the lowest pill burden (1 pill once daily).

Patients initiating antiretroviral therapy should be willing and able to commit to lifelong treatment and should understand the benefits and risks of therapy and the importance of adherence.

CASE TWO

Short case number: 4_20_02

Category: Immune & Haemopoetic Systems

Discipline: Medicine

Setting: Hospital_Urban

Topic: Hodgkin Lymphoma

CASE

You are a medical intern admitting Jason Atkins who is a 17 year old student who developed a cough over the last month, but had otherwise been well.



The cough persisted so his parents encouraged him to see the family GP, because it was starting to interfere with his studies.

The GP detected cervical lymphadenopathy on the left, with painless nodes, with a 'rubbery consistency'. No other lymph nodes were detected, there was no hepatosplenomegaly. There has been no fever or rash. Jason has already undergone some preliminary investigations including chest X-ray and a cervical lymph node biopsy result.

His chest X-ray demonstrated mediastinal lymphadenopathy and the pathology result confirms Hodgkin Lymphoma.

QUESTIONS

1. What are the differential diagnoses for cervical lymphadenopathy? Describe the histological finding shown.
2. After a series of tests Jason's diagnosis was: Hodgkin Lymphoma Stage III. As the intern, it is your responsibility to make sure the patient understands what the tests are for. How would you explain them?
3. Jason went into remission after ABVD chemotherapy. HL is often portrayed as a model for successful treatment of malignancy. What are some features that contribute to the very favourable outcomes?
4. As HL is highly radiosensitive, combined modality therapy is commonly used. How would you outline to the patient what radiotherapy involves and why it is advantageous to use chemo and radiotherapy together?

ANSWERS

Question 1

What are the differential diagnoses for cervical lymphadenopathy? Describe the histological finding shown.

Table 9.17 Cervical lymph node enlargement – differential diagnosis

Infections	Primary lymph node malignancies
Acute	Hodgkin's lymphoma
Pyogenic infections	Non-Hodgkin's lymphoma
Infective mononucleosis	Chronic lymphocytic leukaemia
Toxoplasmosis	Acute lymphoblastic leukaemia
Cytomegalovirus infection	Secondary malignancies
Infected eczema	Nasopharyngeal
Cat scratch fever	Thyroid
Acute childhood exanthema	Laryngeal
Chronic	Lung
Tuberculosis	Breast
Syphilis	Stomach
Sarcoidosis	Miscellaneous
HIV infection	Kawasaki's syndrome
Autoimmune rheumatic disease	Kikuchi's disease (histiocytic necrotizing lymphadenitis)
Rheumatoid arthritis	
Drug reactions	
Phenytoin	

Pathology: The image shows a large multinucleated or multilobated cell with inclusion body-like nucleoli surrounded by a halo of clear nucleoplasm known as the **Reed–Sternberg (R-S) cell**. The hallmark of HL is the presence of the R-S cell in a lymph node biopsy against a background of a mixed inflammatory cell infiltrate. R-S cells are usually derived from germinal centre B cells

Question 2

After a series of tests Jason's diagnosis was: HL Stage III. As the intern, it is your responsibility to make sure the patient understands what the tests are for. How would you explain them?

a) Initial patient assessment

When a confident diagnosis of HL is made after an adequate biopsy, a comprehensive assessment of the patient is essential, including a detailed history and examination.

- Careful recording of the size and distribution of all visible and/or palpable lesions is essential.
- **'B' symptoms: weight loss >10% in the past 6 mths, fevers over 38°C/drenching night-sweats,** should be specifically asked about.
- Co-morbid conditions, such as heart disease, which could influence the tolerance of treatment, should also be recorded.

b) Blood studies

Routine haematological and biochemical indices (FBC & ESR, LDH, LFTs, albumin and creatinine, which either document organ function or provide prognostic information. Thyroid function tests should be performed if the thyroid region is to be irradiated.

c) Staging and distribution of disease

The Ann Arbor system of staging (see Table 11.1) was developed to characterise the extent of disease in patients with HL, but is also applied to other lymphomas.

Table 11.1 Ann Arbor Staging System

Stage	Distribution of Disease
I	Involvement of a single lymph node region (I) or involvement of a single extralymphatic organ or site (IE)
II	Involvement of two or more lymph node regions on the same side of diaphragm alone (II) or with involvement of contiguous extralymphatic organ or tissue (IIE)
III	Involvement of lymph node regions on both sides of the diaphragm (III), which may include the spleen (IIIS) and/or limited contiguous extralymphatic organ or site (IIIE, IIIES)
IV	Multiple or disseminated foci of involvement of one or more extralymphatic organs or tissues with or without lymphatic involvement

The Ann Arbor system was modified at the Cotswolds meeting to include definitions of bulky disease in the CT era, definition of CT criteria for splenic and liver involvement, and definition of a new category of treatment response with persistent radiological abnormalities of uncertain significance.

Staging procedures

Guideline - Hodgkin lymphoma - staging procedures	Level of Evidence
All patients should undergo CT scans of at least the neck, chest, abdomen & pelvis	IV
Bone marrow biopsy is recommended in at least those cases with stage >IIA	IV
FDG-PET scanning or, if unavailable, gallium scanning, are recommended for staging in all cases. Positron emission tomography (PET) is superior to gallium	IV

d) RADIOLOGY:

- **CXR:** Chest radiography alone is inadequate to stage the thorax.
- **CT scanning:** All patients should undergo at least CT scanning from neck to pelvis. If any site is involved beyond neck to pelvis, it is recommended that baseline CT or MRI studies of the site are performed before commencing therapy, both to facilitate response assessment and to assist in planning radiotherapy if appropriate
- **Functional imaging in staging:** Cross-sectional structural imaging modalities, such as CT scanning and MRI, are capable of evaluating lymph node size. Functional imaging can help distinguish benign from malignant nodes and may image disease in the spleen and other organs that is undetected on CT. Scanning with gallium and positron emission tomography (PET) using the radiopharmaceutical F-18 fluorodeoxyglucose (FDG) have both been used in an effort to increase the accuracy of staging in HL. PET is recommended in preference to gallium scanning for staging in HL. When a functional imaging result is equivocal or may change the treatment strategy, biopsy confirmation may be required.

e) Organ function studies

Baseline lung function tests are recommended if bleomycin or thoracic radiation are contemplated. If anthracycline-based chemotherapy is to be given, baseline measurement of left ventricular function is recommended.

f) Bone marrow examination

Bone marrow aspiration and trephine biopsy have a relatively low yield overall in HL. The incidence of bone marrow involvement was 5% in the German HD4-6 study generation, which included 2307 patients in all stages. The marrow positivity rate is particularly low, less than 1%, in patients without B symptoms and with otherwise stage I-II disease.³⁰ Nevertheless, despite the extremely low yield in patients with apparently early-stage disease, the procedure is safe and if positive, has a profound impact on the management of the patients with otherwise early-stage disease. Therefore it can be considered even in these cases.

Question 3

Jason went into remission after ABVD chemotherapy. HL is often portrayed as a model for successful treatment of malignancy. What are some features that contribute to the very favourable outcomes?

HL is one of the best-characterised malignancies of the lymphatic system and one of the forms of malignant disease most readily curable by radiotherapy, chemotherapy or a combination of the two. Modern treatment methods routinely achieve such high cure rates that a very strong emphasis is now placed on achieving cure with the least possible risk of complications from therapy.

- HL is highly radiosensitive. Curative doses of radiation can generally be delivered that are well within normal tissue tolerances.
- HL is one of the malignancies most sensitive to chemotherapy. Combination chemotherapy is curative in > 70% of patients with advanced-stage HL and can produce cure rates of more than 90% when combined with radiotherapy in patients with early-stage disease.

The most commonly used chemotherapy regimens include:

MOPP: mechlorethamine, vincristine, procarbazine and prednisolone

ABVD: doxorubicin, bleomycin, vinblastine and dacarbazine

BEACOPP: bleomycin, etoposide, doxorubicin, cyclophosphamide, vincristine, procarbazine, and Prednisone

Stanford V doxorubicin, vinblastine, mechlorethamine, etoposide, vincristine, bleomycin, prednisone.

Combined-modality therapy is now used for the majority of patients with early-stage disease and is recommended for all patients with bulky mediastinal masses. The use of combined chemotherapy and radiotherapy offers the potential for both reduced toxicity and superior freedom from progression. Fewer cycles of chemotherapy are generally required and radiotherapy is made less toxic by the use of lower doses combined with smaller radiation fields. When chemotherapy and radiotherapy are given together as components of combined-modality therapy, chemotherapy is generally administered first and to the full intended doses. Radiotherapy is commenced after enough time has elapsed to allow haematological recovery, typically 2-4 weeks.

Note that surgery has no place in the primary treatment of HL, but may play a crucial role in obtaining adequate biopsy material for diagnosis, in staging under special circumstances, and in the assessment of residual masses after therapy.

Question 4

As HL is highly radiosensitive, combined modality therapy is commonly used. How would you outline to the patient what radiotherapy involves and why it is advantageous to use chemo and radiotherapy together?

Radiotherapy for HL is a local treatment used to consolidate the chemotherapy treatment. It uses external beam radiation to sterilise residual malignant cells. Patient is not radioactive afterwards. It is delivered in a number of treatments called "fractions", usually Monday to Friday, for 3-5 weeks (depending on the prescribed dose).

The only general side-effect is lethargy which is temporary. It causes lymphopenia (as lymphocytes are exquisitely sensitive to radiation) but rarely clinically significant in small fields. Local side-effects depend on site of treatment and can be acute or late. In this case because of the cervical LN involvement there may be mild odynophagia from oesophagitis following radiotherapy. There are also late side-effects to be considered e.g. hypothyroidism and risk of secondary malignancies.

It is advantageous to use chemotherapy and radiotherapy together due to spatial co-operation: radiotherapy targets local disease whereas chemotherapy also treats systemically. Different toxicities mean that therapeutic ratio is increased compared to either modality being used alone. I.e. lower doses of radiation can be delivered, and with fewer number of cycles of chemotherapy. In other solid malignancies, chemotherapy can be also given concurrently to increase the effectiveness of radiotherapy i.e. radiosensitisation.

CASE THREE

Short case number: 4_20_03

Category: Immune & Haemopoetic Systems

Discipline: Medicine

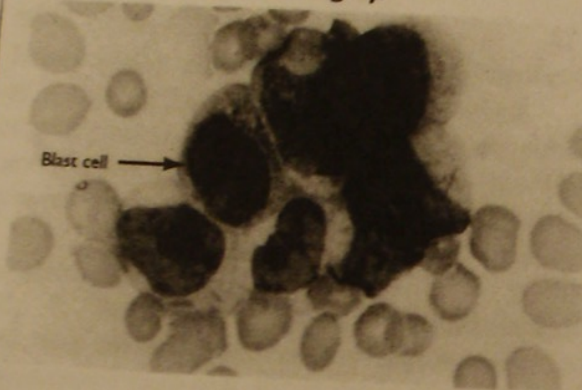
Setting: General practice

Topic: Acute myeloid leukaemia (with BMT)

CASE

Sophie Carpenter is a 50 year-old hairdresser who presents to you with a history of recent onset of menorrhagia for the past few days. She has also noticed that although everyone in her family recently had the flu 3 weeks ago, she seemed to be the only one with lingering symptoms of tiredness and occasional night sweats.

On examination, she was afebrile and you detect pallor of the conjunctiva and some bruises on her arms and lower legs. There was no lymphadenopathy, abdominal tenderness or hepatomegaly/splenomegaly.



The results of the urgent FBC are:

Haemoglobin: 9.8 g/l

WCC: $4.5 \times 10^9/l$

Differential:

- Neutrophils $1 \times 10^9/l$
- Lymphocytes $1.5 \times 10^9/l$
- Monocytes $0.5 \times 10^9/l$
- Eosinophils $0.5 \times 10^9/l$
- Blasts $1 \times 10^9/l$

Platelets: $50 \times 10^9/l$

Peripheral blood film as shown.

After explaining to her that you suspect that her tests are highly suspicious of a serious blood disease, you refer Sophie immediately to the haematologist at the nearest teaching hospital for work up and confirmation.

QUESTIONS

1. Revise the underlying basis for the clinical presentation of Acute leukaemias. Contrast the median age at presentation of AML vs ALL (Note: Paediatric case of ALL covered in Med3000 Week 28 Short case 6).
2. Sophie's bone marrow biopsy confirms AML. What tests are used for confirmation of diagnosis? She was offered bone marrow transplantation (BMT) because the type of AML that she has was considered high risk. What types of tests are used to determine risk of poor survival outcomes in acute leukaemia?
3. The patient wants to know a) if she could pass on the cancer to her children and b) what types of treatment she is likely to undergo if her diagnosis is confirmed. You tell her about the Leukaemia Foundation website. How would you explain briefly the Induction, Consolidation and Maintenance phases in therapy?
4. Describe the 2 main categories of BMT. What are "stem cells"? Explain the difference between "graft vs leukaemia" [benefits] and "graft vs host disease" (GVHD) [adverse effects]

ANSWERS

Question 1

Revise the underlying basis for the clinical presentation of Acute leukaemias. Contrast the median age at presentation of AML vs ALL (Note: Paediatric case of ALL covered in Med3000 Week 28 Short case 6).

Clinical presentation of Acute leukaemias

The majority of patients with acute leukaemia present with symptoms reflecting inadequate haematopoiesis secondary to leukaemic cells infiltration of the bone marrow:

- anaemia – shortness of breath on effort, excessive tiredness, weakness
- leucopenia – recurrent infections
- thrombocytopenia – bleeding and bruising
- marrow infiltration – bone pain.

Examination may include:

- pallor
- fever due to infection
- petechiae, purpura, bruises, fundal haemorrhage
- lymphadenopathy, hepatosplenomegaly

Age of presentation: Acute myeloid leukaemia (AML) has a median age at presentation of 65 years and in contrast acute lymphoblastic leukaemia (ALL) has a substantially lower median age at presentation and is the commonest malignancy in childhood.

Question 2

Sophie's bone marrow biopsy confirms AML. What tests are used for confirmation of diagnosis? She was offered bone marrow transplantation (BMT) because the type of AML that she has was considered high risk. What types of tests are used to determine risk of poor survival outcomes in acute leukaemia?

Confirmation of diagnosis:

- **Blood count.** Hb low, WBC raised usually (sometimes low), platelets low.
- **Blood film.** Blast cells almost invariably seen, lineage identified morphologically, confirmed with immunophenotyping.
- **Bone marrow aspirate.** Increased cellularity, reduced erythropoiesis, reduced megakaryocytes, sometimes trilineage dysplasia. Replacement by blast cells > 20% (often approaching 100%)

Further tests in AML which help in categorizing of risk of poor survival outcome:

- Immunophenotyping e.g. FISH
- cytogenetics
- molecular genetics
- HLA typing as availability of a suitable matched donor would be another driver in decision to include BMT.

Risk of poor survival outcome is mainly based on the cytogenetic pattern and this will help determine treatment options:

- Those at 'low risk' are treated with moderately intensive combination chemotherapy. For those with a low risk profile, allogeneic transplantation is not recommended to consolidate first remission. In this group the potential morbidity from the procedure outweighs the potential risk of relapse.
- Those at 'high risk' may only be treated with curative intent if an HLA-identified sibling is available for BMT.
- Those at 'intermediate risk' should be given consolidating chemotherapy to induce remission followed by matched allogeneic transplantation, despite its attendant risks.

Question 3

The patient wants to know a) if she could pass on the cancer to her children and b) what types of treatment she is likely to undergo if her diagnosis is confirmed. You tell her about the Leukaemia Foundation website. How would you explain briefly the Induction, Consolidation and Maintenance phases in therapy?

a) You reassure her that she will not pass AML to her children in future and you either refer her to <https://www.eviq.org.au/> section on "Cancer Genetics" or print the relevant information for her off this Cancer Institute of NSW website.

b) http://www.leukaemia.com.au/web/aboutdiseases/leukaemias_aml.php

Treatment with curative intent is undertaken in the majority of adults <60 years, provided there is no significant co-morbidity. Complete remission will be achieved in about three-quarters of patients < 60, failure being due to either resistant leukaemia or death due to infection.

- **Induction** of remission: The initial requirement of therapy is to return the peripheral blood and bone marrow to normal i.e. complete remission, using a combination of highly toxic drugs such (Daunorubicin, Cytarabine and Etoposide in AML). The patient goes through a period of severe bone marrow hypoplasia and is at risk of major risk of life-threatening infection in the short term.
- **Consolidation** of remission: Successful remission induction is always followed by further treatment because recurrence is almost invariable if 'consolidation' therapy is not given. This consists of a number of courses of chemotherapy again resulting in periods of bone marrow hypoplasia. BMT could be included in this phase.
- **Maintenance** of remission: Not considered to be of benefit to AML patients who have been brought into complete remission by the previous 2 phases.

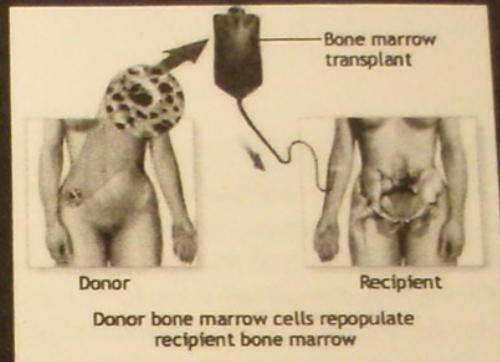
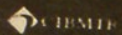
Question 4

Describe the 2 main categories of BMT. What are "stem cells"? Explain the difference between "graft vs leukaemia" and "graft vs host disease" (GVHD).

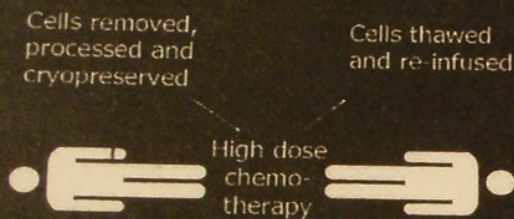
- **Autologous:** In this procedure the patient's own stem cells are first harvested and frozen. After conditioning therapy the stem cells are reinfused in order to repopulate the bone marrow which has become aplastic following chemotherapy. There is no risk of GVHD and no immunosuppression is required. Thus autologous stem cell transplant carries a lower mortality rate of 5% but there is a higher relapse rate.
- **Allogeneic:** Healthy stem cells from a donor are infused intravenously into the recipient who has undergone "conditioning". **The conditioning treatment (chemotherapy +/- radiotherapy) destroys malignant cells and immunosuppresses the recipient as well as ablating the recipient's haematopoietic tissues.** The injected donor stem cells 'home' to

the bone marrow, engraft and produce enough RBCs, granulocytes and platelets after 3-4 weeks. In that window period the patient is at high risk of infection and bleeding and requires intensive supportive care.

Transplant activity worldwide 1980-2009



Autologous transplantation



Allogeneic transplantation



High-dose
chemo-therapy
± radiotherapy

Cells removed
(processed)
then infused



Stem cells are undeveloped cells capable of proliferation, self-renewal, conversion to differentiated cells and regenerating tissues. The sources of stem cells are:

- Peripheral blood
- Bone marrow
- Cord blood

Most bone marrow transplants are now done via the technique of Peripheral Blood Stem Cell Transplant (PBSCT). In this technique, the donor of the blood stem cells is treated with a drug (e.g. the cytotoxic agent cyclophosphamide or the growth factor filgrastim) which encourages release of haematopoietic stem cells from the bone marrow into the peripheral blood. When high levels of the stem cells are detected in peripheral blood in the next few days, these stem cells can be "harvested" (i.e. removed like a normal blood donation), and then "cryopreserved" before re-infusion (like a blood transfusion) into the patient. Once the stem cells are re-infused, they "home" to the bone marrow, and start growing to re-form the patient's red and white blood cells and platelets. This process typically takes 2-4 weeks during which time the patient is at high risk of infection and bleeding, because their own white blood and platelet counts are low.

Total body irradiation: Radiotherapy is given to the whole body to achieve immunosuppression (lymphocyte cell kill) to allow engraftment of donor marrow. It is usually delivered just before the transplant in 3 consecutive days with twice daily treatment.

Graft vs leukaemia: This occurs in the case of an allogeneic stem cell transplant where the healthy donor stem cells are immunologically intact and are able to recognize residual malignant cells in the patient (left over from the conditioning therapy) and destroy them.

Graft vs host disease: GVHD is due to the cytotoxic activity of donor's T lymphocytes becoming sensitized to the host/patient and resulting in an acute or chronic inflammatory reaction. Sites mainly affected are: skin -> rashes, liver -> jaundice, gut -> diarrhea. This can be minimised by good HLA-matching between donor and recipient and using immunosuppressive drugs. The chronic form resembles a connective tissue disorder and is usually treated with corticosteroids and prolonged immunosuppression.

CASE FOUR

Short case number: 4_20_04

Category: Immune & Haemopoetic Systems

Discipline: Medicine

Setting: General practice

Topic: Thalassemia carrier

CASE



Maria Kotsiou is a 22 year old Greek woman who presents to you for vitamins because she has been getting very tired recently. At first she put it down to the stresses of planning for her wedding in 6 months' time but lately it has not improved even with rest.

Examination was normal and, in particular, there was no pallor, hepatosplenomegaly, lymphadenopathy or other physical findings. The investigations are all normal with the exception of the FBC and peripheral smear as shown:

Haemoglobin 11 g/dl (N 11.5-16 Female)

Mean corpuscular haemoglobin MCH 24 pg/cell (N 27-32)

Mean corpuscular Hb concentration MCHC 290 g/l (N 320-360)

Mean corpuscular volume MCV 70 fl (N 80-96)

Reticulocytes 1% (N 0.5-2.5%)

QUESTIONS

1. What does the FBC show? What is the main differential diagnosis and what further tests would help you to confirm your diagnosis?
2. The provisional diagnosis that you make is β -thalassemia trait. How would you explain this condition to Maria? What further tests would confirm your diagnosis?
3. Maria told you that she and her fiancé Kon, whose family is from the same village in Greece as hers, are very keen to have children. Outline what key points about β -thalassemia carrier state you would discuss with her and what services could you refer her to.

Resources:

- Kumar P & Clark M. Clinical Medicine, 7th edition. Chapter 8 Haematological disease. Pp 407-409. Saunders Elsevier 2009.
- Craig JIO, McClelland DBL and Ludlam CA. Blood Disorders: Anemias, Haemoglobinopathies. Chap 24, Davidson's Principles and Practice of Medicine 20th Edition. Churchill Livingstone Elsevier 2006. Pp 1035-38.
- Webster BH and Lammi AT. Thalassaemia and other haemoglobinopathies in general practice. Australian Family Physician 1994 Aug; 23(8):1485-90.

ANSWERS

1. Using the table of normal values from K&C below, what does the FBC show? What is the main differential diagnosis and what further tests would help you to confirm your diagnosis?

Maria's blood picture is diagnostic of microcytic hypochromic anaemia with "target cells" shown in the peripheral blood film. This is based on:

- Haemoglobin < 11.5 g/dl in an adult female
- Microcytic features are MCV < 76 fl (normal 80 – 100)
- Hypochromic features are MCHC

The most common cause of a microcytic hypochromic anaemia is **iron deficiency anaemia**. Other causes are sideroblastic anaemia and the anaemia of chronic disease (often normocytic with the last). **The features of target cells and basophilic stripling are more prominent in haemoglobinopathies like Thalassaemia trait.** The Red cell distribution width (RDW)⁷ is typically increased in iron deficiency anaemia, whereas in Thalassaemia it is usually within normal range.

Other tests which will help in confirming your diagnosis are:

- Iron studies as listed below (serum Fe, TIBC, serum soluble transfer factors, iron in marrow and iron in erythroblasts)

Table 8.3 Microcytic anaemia: the differential diagnosis

	Iron deficiency	Anaemia of chronic disease	Thalassaemia trait (α or β)
MCV	Reduced	Low normal or normal	Very low for degree of anaemia
Serum iron	Reduced	Reduced	Normal
Serum TIBC	Raised	Reduced	Normal
Serum ferritin	Reduced	Normal or raised	Normal
Serum soluble transfer receptors	Increased	Normal	Normal or raised
Iron in marrow	Absent	Present	Present
Iron in erythroblasts	Absent	Absent or reduced	Present

TIBC, total iron binding capacity.

- Haemoglobin electrophoresis
- DNA testing

In summary, the heterozygous β-thalassaemia carrier state patients are usually asymptomatic. Anaemia is mild or absent. The red cells are hypochromic and microcytic with a low MCV and MCH, and it may be confused with iron deficiency. However, the two are easily distinguished, as in **Thalassaemia trait the serum ferritin and the iron stores are normal.** The RDW is usually normal.

Reminder from Med3000 Wk 26 that within the red cells, iron is inserted into the protoporphyrin rings to form haem, and the haem molecule is then inserted into the globin chains to form Hb.

⁷ RDW - The red cell distribution width is calculated by dividing the standard deviation of the red cell width by the mean cell width × 100. An elevated RDW suggests variation in red cell size, i.e. as seen in iron deficiency. In β thalassaemia trait, the RDW is usually normal.

Each normal adult Hb molecule (Hb A) consists of two α and two β globin polypeptide chains. Normally there is balanced (1 : 1) production of α and β chains in globin synthesis. In β -thalassaemia the absent/reduced synthesis of globin chains leads to ineffective erythropoiesis from haemolysis.

HbA comprises about 97% of the Hb in adults. Two other types, Hb A₂ ($\alpha_2\delta_2$) and Hb F ($\alpha_2\gamma_2$), are found in adults in small amounts (1.5–3.2% and < 1%, respectively). In thalassaemia trait Hb electrophoresis usually shows a raised Hb A₂ and often a raised Hb F.

GP's role

- Haemoglobinopathy carrier testing is recommended and should be discussed as part of pre-conceptual/antenatal care in the following individuals:
 - > With a family history of anaemia/thalassaemia/abnormal haemoglobin variant
 - > With any of the following ethnic backgrounds: Southern European, Middle Eastern, African, Chinese, South East Asian, Indian subcontinent, Pacific Islander, New Zealand Maori, South American and some northern Western Australian and Northern Territory Australian Indigenous communities
 - > Partners of known or identified haemoglobinopathy carriers
 - > When MCV < 80fL or MCH < 27pg
- Test partners of pregnant at-risk women at the same time as the pregnant woman where possible.
- Investigate with the following:
 - > FBE
 - > Iron studies (ferritin)
 - > Haemoglobin electrophoresis
- Indications for DNA testing:
 - > Possible carrier for α -thalassaemia (low-borderline MCV or MCH, normal ferritin and normal haemoglobin electrophoresis).
 - > Proven carrier for β -thalassaemia and partner is also a carrier for thalassaemia or other haemoglobinopathy.
 - > Confirmation of carrier status for a haemoglobin variant.
- Where a DNA test is positive, refer to specialist service (haematology clinic, Genetics Services, thalassaemia clinic).
- If both partners of a couple are carriers, refer to Genetics Services and/or haematology clinic for genetic counseling and testing. **This is particularly urgent for pregnant couples (see Contacts, support and testing).**
- Cascade genetic testing is recommended and should be discussed (see Contacts, support and testing).
- Refer the family to a relevant support group (see Contacts, support and testing).

2. The provisional diagnosis that you make is thalassaemia trait. How would you explain this condition to Maria? What further tests would confirm your diagnosis?

You would explain to Maria that according to WHO, globally approx 3% of adults are carriers for thalassaemias. This condition is more common in countries like Southern Europe, Middle East, Africa, China, South-East Asia, India, Pacific Islands, New Zealand (Maoris), South America and some indigenous communities in Australia.

With β -Thalassaemia, there are 3 clinical syndromes:

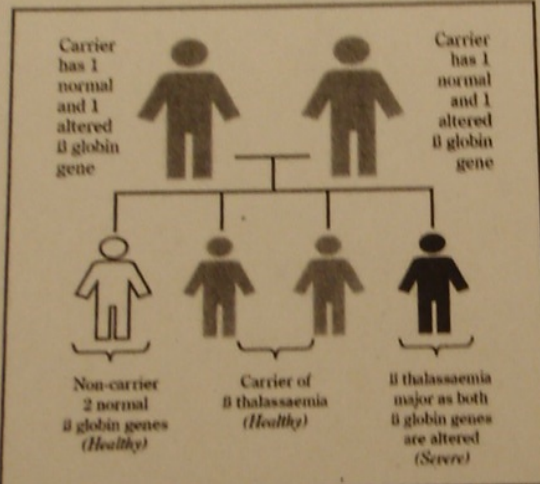
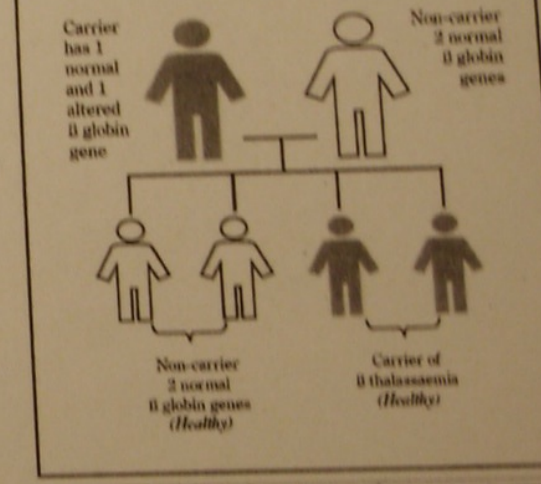
- Thalassaemia minor (or trait), the symptomless heterozygous carrier state
- Thalassaemia intermedia, with moderate anaemia, rarely requiring transfusions
- Thalassaemia major, the homozygous state, which is a severe condition requiring frequent blood transfusions and iron chelation therapy.

You would direct Maria to the β -thalassaemia information on the website:

<http://www.thalassaemia.org.au/thalassaemia-and-related-blood-disorders>

3. Maria is worried that she may not be able to have children. Outline what key you would discuss with her and what services would you refer her to?

Couples planning a family (or early in pregnancy) should have a blood test to determine whether or not they are carriers, **if the origin of either of their families is one of the areas listed**; or if they have a family history of any blood disorder or anaemia. This test is needed to determine if there is any risk of having a child affected by a genetic blood disorder.

If both Maria and Kon are carriers	If Maria is the only parent who is a carrier
 Carrier has 1 normal and 1 altered β globin gene Carrier has 1 normal and 1 altered β globin gene Non-carrier 2 normal β globin genes (Healthy) Carrier of β thalassaemia (Healthy) β thalassaemia major as both β globin genes are altered (Severe)	 Carrier has 1 normal and 1 altered β globin gene Non-carrier 2 normal β globin genes Non-carrier 2 normal β globin genes (Healthy) Carrier of β thalassaemia (Healthy)
With each pregnancy, this couple has a: • 1 in 4 chance of having a child with 2 normal β-globin genes. • 2 in 4 chance of having a carrier of β-thalassaemia. • 1 in 4 chance of β-thalassaemia major.	With each pregnancy, this couple has a: • 2 in 4 chance of having a child with 2 normal β-globin genes. • 2 in 4 chance of having a carrier of β-thalassaemia.

You would either direct Maria to

<http://www.thalassaemia.org.au/thalassaemia-and-related-blood-disorders>

or print off information from the website and go through the relevant material with her.

Figure 2. Suggested protocol for targeted carrier testing of high risk populations¹ for patients in ANY ONE of the categories below:

